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Technical methods of carbon capture by seawater-based solution mineral carbonation

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Abstract

With global industrialization and the excessive use of fossil fuels, greenhouse gas emissions have substantially increased, particularly the concentrations of carbon dioxide (CO_2), which have had profound impacts on global warming and extreme climate events. As a key technology to combat climate change, mineral carbonation has been widely recognized as an effective technique for CO_2 sequestration and utilization. In recent years, seawater-based solutions (including desalination brine and seawater-based wastewater) have become a research hotspot in mineral carbonation due to their strong alkalinity, high content of Ca^{2+} and Mg^{2+}, and lower costs compared to chemical solvents. This article comprehensively reviews the CO_2 mineral carbonation mechanisms based on seawater-based solutions, highlighting recent research achievements and potential development trends in capturing CO_2 through the pH swing method, modified Solvay process, chemical precipitation method, and chemical absorption method. By comparing the use of seawater-based solutions for mineral carbonation with traditional techniques, this study evaluates its advantages and the challenges it may face in practical applications, aiming to provide a more accurate assessment of the impact and future development of this technology.

Key words CO_2 sequestration and utilization, seawater-based solution, mineral carbonation

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1 Introduction

With the development of global industrialization, the excessive use of fossil fuels has led to significant emissions of greenhouse gases such as carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), chlorofluorocarbons (CFCs), and ozone (O_3) (Mysiak et al., 2016). Notably, the rapid increase in atmospheric CO_2 concentrations significantly impacts global warming, leading to more frequent extreme weather events and environmental disasters such as floods, droughts, and rising sea levels (Kapila et al., 2019). The global annual average concentration of atmospheric CO_2 in 2024 reached approximately 423.64×10^{-6}, representing an increase of about 3.7×10^{-6} compared to 2023 (https://gml.noaa.gov/ccgg/trends/global.html). Thus, reducing CO_2 emissions is imperative. On the other hand, global energy demand is continuously rising (Jackson et al., 2018). The projected increase in global energy demand by 28.6% by 2040 is compared to the baseline year 2017 in the 2018 report by the International Energy Agency (IEA, 2019). Reducing CO_2 emissions is challenging in the short term, hence the need for

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increased negative emissions, i.e., the irreversible capture and storage of CO_2 through physical, chemical, or biological means (Liu et al., 2021).

Carbon capture and storage (CCS) is a cornerstone of current emission reduction strategies and is expected to sequester over 7 × 10^9 t of CO_2 annually by 2050. Traditional CCS approaches include geological storage, ocean storage, and mineral storage. While geological storage offers high capacity, it raises concerns about leakage, induced seismicity, and groundwater acidification. Ocean storage—injecting CO_2 into deep waters—may adversely affect marine ecosystems. In contrast, mineral carbonation, also called CO_2 mineral storage, accelerates the natural weathering of rocks, reacting CO_2 with calcium- or magnesium-bearing materials to form stable carbonates such as $CaCO_3$ and $MgCO_3$. This process offers long-term stability without ongoing monitoring and has high industrial potential (Liu et al., 2021). Mineral carbonation is particularly well-suited for small- and medium-scale emitters (<2.5 Mt CO_2), which account for 10%–15% of global emissions (Sanna et al., 2012). Furthermore, carbon distribution in the Earth's lithosphere (Fig. 1) shows that over half exists as carbonate minerals, underscoring the natural stability of this storage form.

Conventional mineral carbonation typically relies on natural silicate rocks (e.g., olivine, serpentine), chemical reagents [e.g., CaO, $Ca(OH)_2$], or industrial alkaline wastes (e.g., blast furnace or steel slag) as feedstock (Kapila et al., 2019; Wang et al., 2023). However, the use of distilled water, acid solvents, and high energy input significantly increases costs and hampers scalability (Giannoulakis et al., 2014; Sanna et al., 2014). In contrast, seawater presents a cost-effective and abundant alternative. With a pH of 8.0–8.2 and naturally high concentrations of Ca^{2+} and Mg^{2+}, it inherently facilitates carbonate mineral formation (Li et al., 2022; Liu et al., 2024). Beyond natural seawater, two seawater-based solutions, desalination brine and seawater-based wastewater, are emerging as ideal media for CO_2 mineralization due to their enriched Ca^{2+} and Mg^{2+} content (Wang et al., 2020; Myers et al., 2022). Desalination brine, produced at 95 × 10^6 m^3/d globally, contains significantly higher Ca^{2+} (0.71–0.80 g/L) and Mg^{2+} (2.11–2.42 g/L) levels than natural seawater (Bang et al., 2019; Zhao et al., 2020). Its reuse not only supports CO_2 sequestration but also addresses the growing challenge of brine disposal in coastal and arid regions (Mustafa et al., 2020). Even more concentrated, seawater-based wastewater from salt and marine chemical production contains Ca^{2+} (21.3–26.9 g/L) and Mg^{2+} (21.0–56.4 g/L) (Ho and Iizuka, 2023). Though potentially hazardous if discharged untreated, it offers a highly reactive medium for converting waste into valuable, stable carbonates (Alkhadra et al., 2020).

The use of seawater-based solutions offers a dual benefit: cost-effective CO_2 capture and the simultaneous treatment of saline industrial effluents, delivering synergistic environmental gains. This approach supports circular economy goals and low-carbon wastewater management. Recent studies have examined various aspects of seawater-based solution CO_2 mineral carbonation, including reactor design, reaction kinetics, product utilization, and techno-economic feasibility (Al-Anezi and Hilal, 2006; Ibrahim et al., 2019; Veetil and Hitch, 2020; Thonemann et al., 2022; Yoo et al., 2022; Choi et al., 2023; Ding et al., 2023; Lee et al., 2024; Liu et al., 2024). Nevertheless, challenges persist, particularly in enhancing reaction efficiency, reducing energy demand, scaling systems, and ensuring long-term environmental safety. This review aims to systematically summarize the principles, development, and current progress of CO_2 mineral carbonation using seawater-based solutions, assess key technical and environmental challenges, and propose future research directions. Emphasis is placed on the dual benefits of this approach in CO_2 sequestration and seawater-based solutions remediation. This strategy offers great promise for scalable deployment, especially for small- and

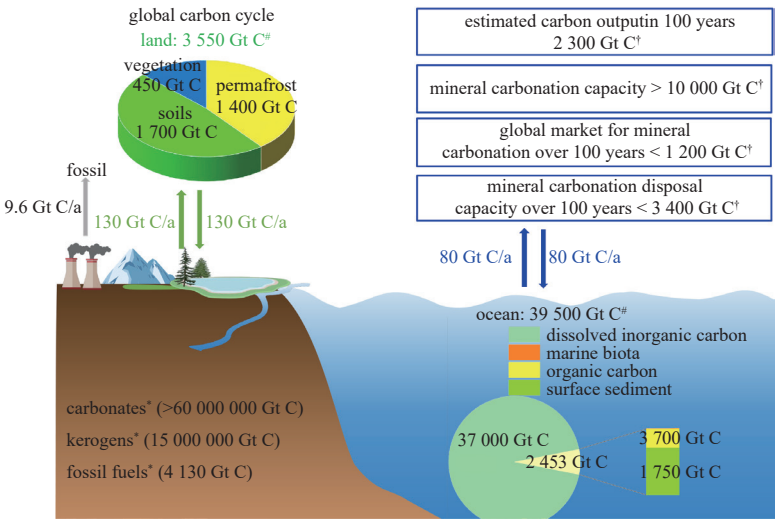

Fig. 1. Global carbon reservoirs (Gt C) and net fluxes (Gt C/a) (* Falkowski et al., 2000; # Friedlingstein et al., 2023; † Sanna et al., 2012) (modified from Sanna et al., 2014 and Liu et al., 2024).

medium-scale CO_2 emitters, and provides an innovative, seawater-based solution for climate change mitigation.

2 Principles and development status of seawater-based solution mineral carbonation technology

Seawater-based solution mineral carbonation provides a sustainable and scalable approach to mitigate atmospheric CO_2 by promoting its transformation into stable carbonate minerals through reactions with Ca^{2+} and Mg^{2+} in aqueous media. This process is typically divided into three sequential steps: (1) alkalization of seawater to enhance CO_2 solubility, (2) conversion of dissolved CO_2 into carbonate (CO_3^{2-}) and bicarbonate (HCO_3^-) species, and (3) precipitation of carbonate solids such as $CaCO_3$ and $MgCO_3$ (Fig. 2, Ho and Iizuka, 2023). Over the past decade, research efforts have evolved into four principal pathways for operational implementation: the pH swing method, the modified Solvay process, the chemical precipitation method, and the chemical absorption method. Each exhibits distinct mechanistic advantages, maturity levels, and integration potential. The following subsections elaborate on the reaction mechanisms, technical advances, and experimental progress of these pathways (Table 1).

2.1 *pH swing method*

The basic principle of capturing CO_2 through mineral carbonation using seawater-based solutions with the pH swing method involves adding alkaline substances to enhance the dissolution of CO_2 and adjusting the pH of the seawater to separate Ca^{2+} and Mg^{2+} ions, thereby obtaining high-purity carbonation products (Fig. 3). This method can achieve pure products because Ca^{2+} and Mg^{2+} can be precipitated by adjusting the pH of the solution separately. For example, a solution devoid of Mg^{2+} and rich in Ca^{2+} can be obtained at pH values ranging from 10.36 to 12.28 (Fig. 3), mineralizing Ca^{2+} alone and forming pure calcium carbonate products (Xie et al., 2015).

Research on the pH swing method primarily focuses on controlling solution pH to facilitate efficient carbonation reactions. For example, Wang et al. (2011) investigated CO_2 mineralization in seawater using a gas mixture containing 15% CO_2 and 85% N_2, with NH_4Cl and NH_3 serving as buffering agents. This allows for the precipitation of Ca^{2+} and Mg^{2+} from seawater as $CaCO_3$, $MgCO_3$, and calcium magnesium carbonate [$CaMg(CO_3)_2$]. Their study estimated that approximately 2.65 kg of CO_2 could be fixed by 1.34 m^3 natural seawater, with desalination brine enhancing this potential by 2–3 times, achieving up to 7.95 g CO_2/L. Building on this concept, Bang et al. (2017) applied a pH swing method involving NaOH to selectively separate Ca^{2+} and Mg^{2+} of desalination brine and promote CO_2 mineralization, thereby enabling cyclic carbonation. Moreover, by rationally designing the pH swing method, competition for precipitation caused by differences in hydration energy between Ca^{2+} and Mg^{2+}, as well as the inhibitory effect of Mg^{2+} on the nucleation and growth of $CaCO_3$, can be mitigated (Zhang et al., 2018; Pan et al., 2021). Under optimized conditions using a 15% CO_2 gas stream, the consumption of NaOH was reduced by 76% while achieving a CO_2 conversion rate of 69% (Bang et al., 2019). Zhang et al. (2019) employed blast furnace slag and NaOH as alkaline sources and introduced nickel nanoparticles (NiNPs) as catalysts to further enhance process practicality and reduce material costs. By controlling the CO_2 dissolution rate, they successfully produced commercial-grade $CaCO_3$, demonstrating the method's industrial adaptability and scalability. Recent research by Lee et al. (2024) demonstrated that, using desalination brine as the reaction medium, pH swing systems based on NaOH significantly outperform those involving monoethanolamine (MEA) addition in terms of carbonation kinetics and carbonate purity. They further found that the Ca^{2+} and CO_3^{2-} ratio has a critical influ-

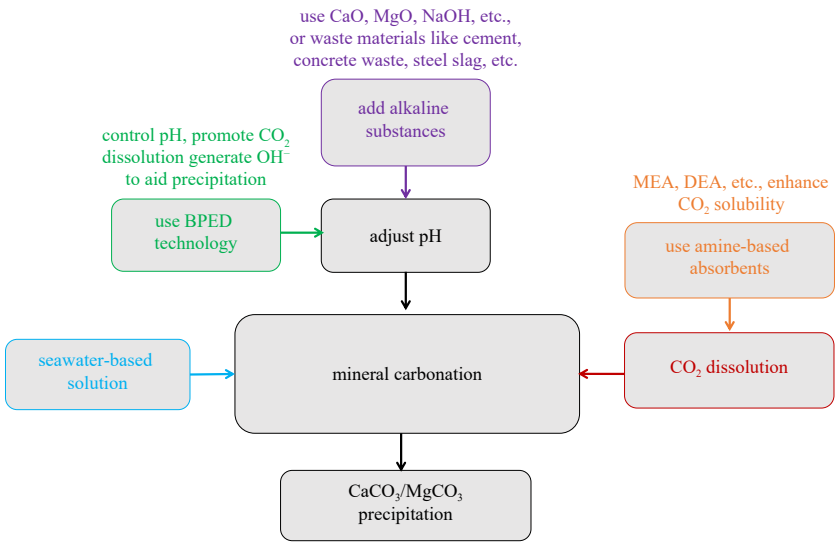

Fig. 2. Pathways and principles of seawater-based solution mineral carbonation for CO_2 capture.

Table 1. Review of CO_2 capture technologies by mineral carbonation of seawater-based solutions (modified from Ho and Iizuka, 2023)

Method	Sample type	pH regulation	Reactant	Mineral product	CO_2 capture performance	Feature/advantage	Reference
pH swing method	desalination brine	NH_4Cl + NH_3 buffer	CO_2 + N_2	$CaCO_3$, $MgCO_3$, $CaMg(CO_3)_2$	CO_2 absorption capacity: 7.95 g CO_2/L brine	sequential precipitation of Ca^{2+} and Mg^{2+}	Wang et al. (2011)
pH swing method	desalination brine	NaOH	CO_2	$CaCO_3$, $Mg_5(CO_3)_4(OH)_2 \cdot 4H_2O$	maximum Ca^{2+} conversion: 99%; maximum Mg^{2+} conversion: 86%; maximum CO_2 absorption capacity: 4.27×10^{-3} t/t brine	control pH to separate Ca^{2+} and Mg^{2+} and enhance CO_2 mineral carbonation	Bang et al. (2017)
pH swing method	desalination brine	NaOH	CO_2	$CaCO_3$, $Mg_5(CO_3)_4(OH)_2 \cdot 4H_2O$	CO_2 capture efficiency: 69%	control pH to separate Ca^{2+} and Mg^{2+} and enhance CO_2 mineral carbonation	Bang et al. (2019)
pH swing method	desalination brine	NaOH + industrial alkaline slag	CO_2 + NiNPs	$CaCO_3$	commercial-grade $CaCO_3$	catalytically regulate CO_2 dissolution rate to optimize mineralization efficiency	Zhang et al. (2019)
pH swing method	desalination brine	NaOH	CO_2	$CaCO_3$	–	Ca^{2+} and CO_3^{2-} ratio affects $CaCO_3$ crystal size and structure; NaOH outperforms MEA in kinetics and carbonate conversion	Lee et al. (2024)
pH swing method	seawater-based wastewater	NaOH	CO_2 + MEA	$MgCO_3 \cdot 3H_2O$, $CaCO_3$	theoretical CO_2 absorption capacity: 0.672 t CO_2/d; maximum product purity: $MgCO_3 \cdot 3H_2O$ (95.15%), $NaHCO_3$ (96.51%)	first combination of amine absorption with Ca^{2+}/Mg^{2+} separation	Yoo et al. (2018)
pH swing method	seawater-based wastewater	NaOH	CO_2 + primary amines	$MgCO_3 \cdot 3H_2O$, $Mg_5(CO_3)_4(OH)_2 \cdot 5H_2O$	the longer the carbon chain, the higher the absorption efficiency	longer amine chains improve absorption	Yoo et al. (2019)
pH swing method	seawater-based wastewater	NaOH	CO_2 + MEA	$CaCO_3$, $MgCO_3$, $NaHCO_3$	maximum CO_2 capture efficiency: 44.3%	formation of three types of mineral products, enhancing synergistic resource utilization	Kang et al. (2020)
pH swing method	seawater-based wastewater	electrolysis (NaOH/HCl)	CO_2	$CaCO_3$, $MgCO_3$	CO_2 capture efficiency: 91.3%; $CaCO_3$ conversion: 90%; $MgCO_3$ conversion: 60%; energy consumption: 16 548 kWh/t CO_2	internal alkali/acid recycling	Choi et al. (2021)
pH swing method	seawater-based wastewater	NaOH	CO_2 + SO_2	$CaCO_3$, $MgCO_3$, gypsum	CO_2 capture efficiency: 91%; desulfurization efficiency: 99%; $CaCO_3$ conversion: 99.9%; $MgCO_3$ conversion: 63%	simultaneous CO_2 capture and desulfurization enhances byproduct utilization and produces commercial-grade gypsum	Cho et al. (2022)

to be continued

continued from Table 1

Method	Sample type	pH regulation	Reactant	Mineral product	CO_2 capture performance	Feature/advantage	Reference
pH swing method	seawater-based wastewater	NaOH	$CO_2 + SO_2$	$Mg_5(CO_3)_4(OH)_2 \cdot 4H_2O$, $Mg_3Ca(CO_3)_4$, $MgCO_3 \cdot 3H_2O$, gypsum	enables sequential absorption: SO_2 is preferentially captured and mineralized; CO_2 is partially absorbed and converted	leverages thermodynamic selectivity of NaOH; achieves simultaneous gas separation and solid byproduct recovery; controls particle morphology via saturation and ionic regulation	Choi et al. (2023)
Modified Solvay process	desalination brine	CaO	CO_2	$NaHCO_3$	maximum CO_2 capture efficiency: >90%	the solubility of $NaHCO_3$ plays a key role throughout the mineral carbonation process.	El-Naas et al., (2010)
Modified Solvay process	desalination brine	CaO	CO_2 + MEA/AMP/ MDEA/PZ	$NaHCO_3$	maximum CO_2 absorption capacity: more than 0.9 mol CO_2/mol-amine; maximum Na removal: 85%	AMP promotes carboxylation; the maximum $NaHCO_3$ yield reaches 13 g /100 g solution	Dindi et al. (2015)
Modified Solvay process	desalination brine	CaO	CO_2	$NaHCO_3$	CO_2 capture efficiency: 99%; Na^+ removal: 35%; energy consumption: 2.36 MJ/kg CO_2	enhanced alkalinity of brine; superior CO_2 capture and sodium removal performance; optimal at 20℃ and pH >10; reduced energy consumption	El-Naas et al. (2017)
Modified Solvay process	desalination brine	Mg–Al–O	CO_2	$NaHCO_3$	maximum CO_2 absorption capacity: 1.86 mol CO_2/kg solution; maximum $NaHCO_3$ yield: 44 g/kg solution	the adsorbent is regenerable and produces Cl_2 and HCl as byproducts	Dindi et al. (2018)
Modified Solvay process	desalination brine	LiOH	CO_2 + 3-amino-1-propanol	$NaHCO_3$, $CaCO_3$, $MgCO_3$ $3H_2O$	maximum CO_2 absorption capacity: 6.25×10^{-3} t/m^3; Na^+ removal efficiency: 45%	LiOH precipitates $Mg(OH)_2$ and $Ca(OH)_2$. $Mg(OH)_2$ is then used for CO_2 mineralization, and $Ca(OH)_2$ to produce $NaHCO_3$	Lee et al. (2021)
Modified Solvay process	desalination brine	KOH	CO_2	$NaHCO_3$, $KHCO_3$, KCl	maximum CO_2 absorption capacity: 108.2 g CO_2/L brine	adding NH_4HCO_3 and extra KOH lowers the solubility of $NaHCO_3$ and prevents pH reduction, thus enhancing CO_2 absorption	Mourad et al. (2022)
Modified Solvay process	desalination brine	$MgCl_2 \cdot nH_2O$	CO_2	$MgCO_3 \cdot 3H_2O$	no consumable chemicals; 89% Mg^{2+} recovery; 66-95% HCl regeneration; supports zero liquid discharge and net-negative CO_2 emissions	the achieved separation and purification of Mg-chloride makes it possible to use desalination brine as a magnesium source for CO_2 mineralization as magnesium carbonate	Myers et al. (2022)

to be continued

continued from Table 1

Method	Sample type	pH regulation	Reactant	Mineral product	CO_2 capture performance	Feature/advantage	Reference
Chemical precipitation method	seawater-based wastewater	NaOH	CO_2	$MgCO_3$ $3H_2O$, $NaHCO_3$, $CaCO_3$	maximum CO_2 capture efficiency: 92.0%; maximum product purity: $MgCO_3 \cdot 3H_2O$ (93.0%), $NaHCO_3$ (92.2%)	Na_2SO_4 facilitates the selective precipitation of Ca^{2+} as $CaSO_4$; product purity exceeded 92%	Kim et al. (2020)
Chemical precipitation method	seawater-based wastewater	NaOH	CO_2 + Na_2SO_4 + MEA/ DEA/MDEA (30 wt%)	$CaCO_3$, $MgCO_3$	maximum CO_2 capture efficiency: 55.18%; maximum product purity: $CaCO_3$ (95.2%)	product purity exceeded 95.2%; the type of absorbent significantly influenced the efficiency	Kang et al. (2018)
Chemical precipitation method	seawater-based wastewater	NaOH	CO_2	$CaCO_3$, $MgCO_3$, $NaHCO_3$	CO_2 absorption capacity: 48.17 kg CO_2/t wastewater	three carbonate products were generated, with a total yield of 98.15 kg/t of wastewater	Yoo et al. (2022)
Chemical absorption method	seawater-based wastewater	KOH	CO_2 + tributylamine	$CaCO_3$	maximum CO_2 capture efficiency: 95.0%	produces high-purity $CaCO_3$; tributylamine is regenerable with $Ca(OH)_2$ and remains stable after 11 cycles, suitable for commercial filler applications	Liu et al. (2014)
Chemical absorption method	desalination brine	self-buffering capacity of brine	CO_2 + insoluble amine (diisobutylamine) + n-octanol + regeneration with carbide slag	$CaCO_3$, $MgCO_3$	Ca^{2+}conversion: 92%; Mg^{2+} conversion: 80%	a regenerative amine system enables rapid, one-step CO_2 capture and mineralization, resulting in high-value carbonate production, and is compatible with seawater desalination	Wang et al. (2015)
Chemical absorption method	desalination brine	self-buffering capacity of brine	CO_2 + DEA	inorganic metal carbonates	optimal CO_2 capture achieved with 30 wt% DEA, yielding 0.419 3 mol/mol CO_2 loading	DEA-enabled CO_2 capture and mineralization using industrial wastewater shows high efficiency and strong engineering potential	Kang et al. (2017)

Note: MDEA, methyldiethanolamine; DEA, diethanolamine; PZ, piperazine. − denotes that corresponding data were not available.

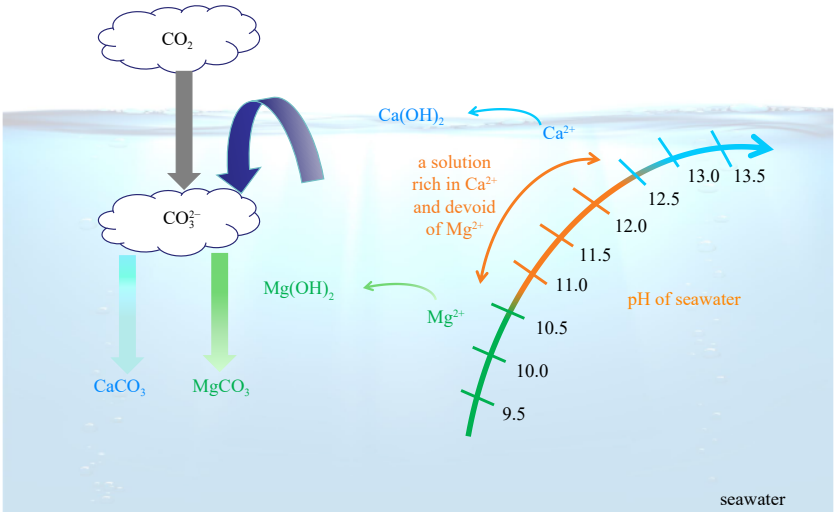

Fig. 3. Separation of Ca^{2+} and Mg^{2+} by pH adjustment (modified from Ho and Iizuka, 2023).

ence on the crystal morphology of $CaCO_3$, highlighting the importance of precise ionic control in optimizing mineral carbonation performance.

The pH swing method has also been widely applied in CO_2 mineralization systems based on seawater-based wastewater. For instance, Yoo et al. (2018) used NaOH to separate Ca^{2+} and Mg^{2+} from seawater-based wastewater, reacting with CO_2 captured by alkanolamine absorbents to form high-purity $CaCO_3$ and $MgCO_3$. However, compared to the precipitation of $CaCO_3$ from $Ca(OH)_2$, precipitating $MgCO_3$ from $Mg(OH)_2$ is more challenging. To address this, Yoo et al. (2019) further investigated the effects of primary alkanolamine alkyl chain length, reaction time, CO_2 concentration, and Mg/Na ratio on the formation efficiency of $MgCO_3$. The results indicated that longer alkyl chains can enhance the CO_2 absorption rate, but the overall improvement was limited due to poor desorption and reabsorption performance. Building upon this, Kang et al. (2020) proposed a more integrated approach by combining CO_2 capture using MEA with the pH swing method to generate $Ca(OH)_2$ and $Mg(OH)_2$, which then reacted to form $CaCO_3$ and $MgCO_3$. The remaining saline solution subsequently reacted with CO_2 to produce $NaHCO_3$. Carbon balance analysis revealed that the contribution of $NaHCO_3$ precipitation to CO_2 uptake surpassed that of $CaCO_3$ and $MgCO_3$, highlighting the potential of this system for synergistic mineralization and resource utilization.

Despite its effectiveness, a key limitation of the pH swing method lies in its high chemical demand, particularly for NaOH. To address this, researchers have integrated electrolysis technologies [e.g., bipolar membrane electrodialysis (BPED) technology] to improve cost-effectiveness and sustainability (Zhao et al., 2020; Choi et al., 2021). Choi et al. (2021) demonstrated that NaOH generated from brine electrolysis can be used to separate Ca^{2+} and Mg^{2+} and facilitate CO_2 dissolution. Under pH 7–11, Mg^{2+} can be selectively precipitated as $Mg(OH)_2$, then electrolytically generated HCl dissolves to recover Mg^{2+} and produce $MgCO_3$ and $CaCO_3$. Notably, this approach allows for internal regeneration of both NaOH and HCl, supporting closed-loop operation. Building on this concept, Cho et al. (2022) further optimized the process for simultaneous CO_2 capture and SO_2 removal. At pH 11–13, residual Ca^{2+} in the brine was precipitated as $Ca(OH)_2$ using NaOH and subsequently reacted with SO_2 to form $CaSO_4$ (desulfurization gypsum), offering an additional pathway for resource recovery and byproduct valorization.

2.2 *Modified Solvay process*

The Solvay process, also known as the ammonia-soda process, is a conventional process used by desalination brine plants to produce sodium carbonate from waste brine, as illustrated in Fig. 4. The equal amount of CO_2 and consuming NH_3, this process does not promote the fixation of CO_2 (Ho and Iizuka, 2023) [Eq. (1)]. To improve this process, El-Naas et al. (2010) and El-Naas et al. (2017) proposed using CaO or $Ca(OH)_2$ instead of NH_3 as the alkali source to produce $NaHCO_3$ as the target product (Fig. 4). This modified Solvay process reduces operational energy and total equivalent power consumption while converting CO_2 into solid bicarbonate [Eq. (2)]. Furthermore, using desalination brine, elements like Ca, Na, and Mg can produce a mix of solid products, including $MgCO_3 \cdot 3H_2O$, $NaHCO_3$, and $CaCO_3$ (Lee et al., 2021).

$$2NaCl + CaCO_3 \leftrightarrow Na_2CO_3 + CaCl_2, \quad (1)$$

$$2NaCl + 2CO_2 + Ca(OH)_2 \leftrightarrow 2NaHCO_3 + CaCl_2 \ (\Delta G_{20°C} = -56.0\ \mathrm{kJ/mol}), \quad (2)$$

where $\Delta G_{20℃}$ represents the change in Gibbs free energy at a temperature of 20°C.

Several studies have investigated the potential of the modified Solvay process for CO_2 capture and resource recovery from desalination brine. Dindi et al. (2015) explored the use of various alkylamines [e.g., MEA, 2-

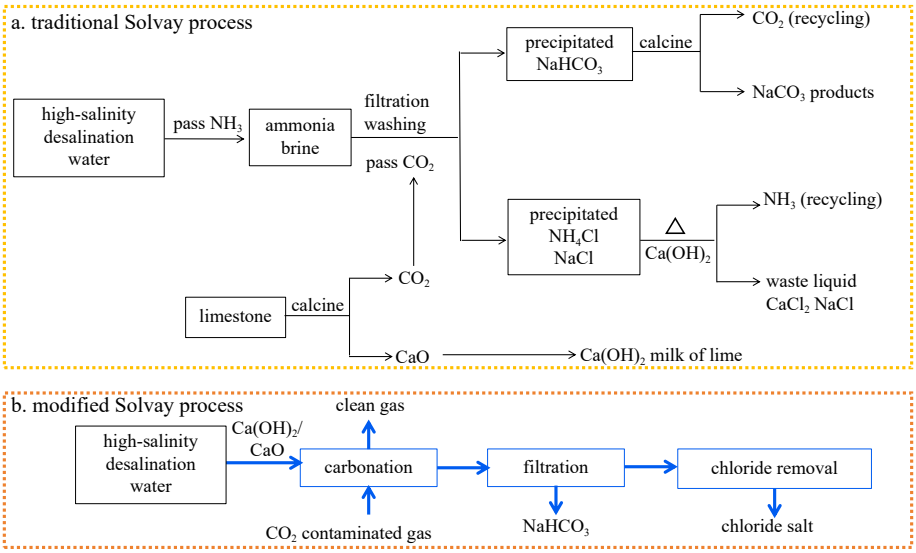

Fig. 4. Schematic diagrams of modified (a) and traditional (b) Solvay processes (modified from El-Naas et al., 2017).

amino-2-methylpropanol (AMP), MDEA, and PZ] and CaO in conjunction with 15% CO_2 to enhance the process performance. Their study reported a maximum $NaHCO_3$ yield of 13 g/100 g solution, with CO_2 absorption exceeding 0.9 mol CO_2/mol amine. Among the amines tested, AMP significantly improved hydrolysis and carbonate precipitation efficiency. El-Naas et al. (2017) employed 10% CO_2 and CaO to treat brines, achieving a CO_2 capture efficiency of up to 99% and a sodium ion removal rate of 35%, with an energy consumption of 2.36 MJ/kg CO_2. This work emphasized the essential role of brine alkalinity in simultaneously enhancing CO_2 uptake and ion exchange. To address the issue of toxic contamination caused by the degradation products of amines in the modified Solvay processes, Dindi et al. (2018) developed a Mg–Al–O composite adsorbent by calcining Mg–Al layered double hydroxide. This material could remove Cl^- from brine while releasing OH^- to promote CO_2 absorption and $NaHCO_3$ formation. Moreover, the adsorbent could be regenerated via calcination, releasing Cl_2 gas and HCl as valuable byproducts, indicating the potential for closed-loop utilization. Building on this foundation, Lee et al. (2021) utilized a modified Solvay process incorporating 15% CO_2, 3-amino-1-propanol, and lithium hydroxide (LiOH) as additives. This configuration achieved a CO_2 absorption capacity of 6.25×10^{-3} t/m^3 and a sodium removal efficiency of 45%. LiOH facilitated the precipitation of $Mg(OH)_2$ and $Ca(OH)_2$, which were subsequently used to mineralize CO_2 and form $NaHCO_3$, highlighting the role of specific alkaline agents in process optimization.

To further enhance CO_2 absorption efficiency and sodium/chloride removal, Mourad et al. (2022) proposed a multistage indirect carbonation strategy. This process incorporated an inert particle fluidized bed reactor and KOH addition, which helped buffer against pH reduction and increase the precipitation rate of $KHCO_3$ [Eq. (3)]. The authors identified KOH concentration, operating pressure, temperature, and CO_2 flow rate as key controlling parameters. Importantly, the system was designed to operate at elevated temperatures (40–55℃), aligning with the thermal discharge conditions of desalination plant effluents (Mourad et al., 2021), which helps improve the solubility of K_2CO_3 and the yield of $KHCO_3$. Additionally, using $MgCl_2 \cdot nH_2O$ derived from desalination brine for CO_2 capture has shown promise. In this pathway, $MgCl_2 \cdot nH_2O$ is thermally decomposed to generate MgO and HCl. The MgO is then reacted with CO_2 in a ball mill carbonator to produce $MgCO_3 \cdot 3H_2O$, while the HCl byproduct is recycled within the system (Wang et al., 2020; Myers et al., 2022). This process integrates thermal treatment with mineral carbonation and offers a practical route for CO_2 capture from industrial-grade saline streams.

$$2NaCl + 2CO_2 + KOH \leftrightarrow NaHCO_3 + KCl \ (\Delta G_{20°C} = -130.6 \text{ kJ/mol}). \quad (3)$$

2.3 *Chemical precipitation method*

The chemical precipitation method enhances selective separation of Ca^{2+} and Mg^{2+} through additive-assisted precipitation, typically using Na_2SO_4 or other salting-out agents. In this route, Ca^{2+} is first precipitated as $CaSO_4$, enabling downstream mineralization of Mg^{2+} with CO_2 (Fig. 5).

The chemical precipitation method for capturing CO_2 through mineral carbonation using seawater-based solutions has made substantial progress. Kim et al. (2020) employed the chemical precipitation method by adding Na_2SO_4 to separate cations from seawater-based wastewater. They designed a two-step indirect mineral carbonation process (Fig. 5), where cation separation is the first step, and $MgCO_3$ precipitation is the second. The precipitation of $CaCO_3$ affects the precipitation of $MgCO_3$, therefore, Na_2SO_4 is used in the first step to separate Ca^{2+} and Mg^{2+}. The separated Ca^{2+} precipitates as $CaSO_4$. MEA

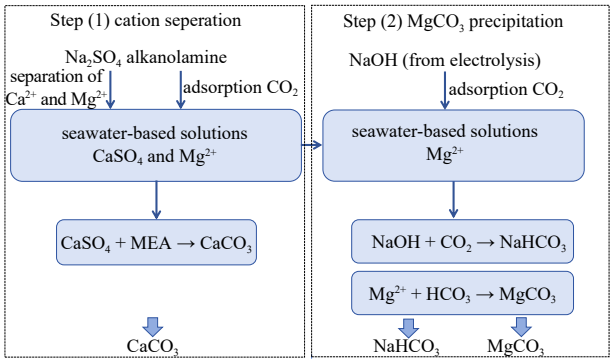

Fig. 5. Schematic diagram of CO_2 capture using seawater-based solution mineral carbonation based on the chemical precipitation method (modified from Kim et al., 2020).

and NaOH are used as absorbents to capture CO_2. The resulting $CaSO_4$ and the remaining solution then react separately with the absorbed CO_2, producing $CaCO_3$ and $MgCO_3$. Considering that NaOH can be obtained through seawater electrolysis, this technique is both economical and practical, effectively converting CO_2 into high-purity metal carbonates. In a similar study, Kang et al. (2018) utilized Na_2SO_4 to precipitate Ca^{2+} from seawater-based industrial wastewater, resulting in the formation of insoluble $CaSO_4$ solids. They then utilized three different adsorbents, MEA, DEA, and MDEA (all at typical industrial concentrations of 30 wt%), to generate high-purity calcium carbonate salts, which existed as a mixture of aragonite and calcite. The conversion rates were 33.16%, 33.86%, and 55.18%, respectively. Thermogravimetric analysis showed that the purity of the products was 93.6%, 95.2%, and 92.8%, respectively. Moreover, although the method requires the addition of NaOH to raise the initial pH and separate Mg^{2+}, the study found that adding Na_2SO_4 further enhanced the selective separation of Ca^{2+}, significantly improving the purity of the final carbonated product. The analysis revealed that the adsorbent influences the mineral carbonation reactions involving metal cations in seawater. Therefore, the study suggests that to improve the efficiency of seawater mineral carbonation, not only should the extraction of metal ions be considered, but the optimization of the adsorbent type should also be studied. The research further indicates that when using 1 L of seawater-based wastewater solution for conversion and precipitation, 1.094 7 mol of CO_2 can be removed. Scaling up this data shows that approximately 1 t of concentrated seawater-based industrial wastewater can remove 48.17 kg of CO_2, generating 38.19 kg of $CaCO_3$, 21.49 kg of $MgCO_3$, and 38.47 kg of $NaHCO_3$. In other words, 1 t of seawater-based wastewater can produce 98.15 kg of metal carbonate materials (Yoo et al., 2022).

2.4 *Chemical absorption method*

Rather than functioning as an independent pathway, the chemical absorption method is most effectively utilized as an auxiliary strategy integrated within other seawater-based mineral carbonation approaches, such as the pH swing method, the modified Solvay process, and chemical precipitation method. In this context, gaseous CO_2 is first captured using alkanolamine-based absorbents — such as MEA, AMP, MDEA, and PZ—which react with CO_2 to form carbamate or bicarbonate species. These products subsequently undergo mineralization with Ca^{2+} and Mg^{2+} present in seawater or brine-derived solutions, leading to the formation of solid carbonates (Huang et al., 2007; Padurean et al., 2012; Kang et al., 2017; Liu et al., 2014).

The integration of chemical absorption enhances the CO_2 capture capacity and selectivity of hybrid systems. For instance, amines facilitate initial CO_2 absorption in pH swing method, after which NaOH or other alkaline additives adjust the solution pH to trigger selective precipitation of $CaCO_3$ and $MgCO_3$ (Yoo et al., 2018). Likewise, in modified Solvay process, alkanolamines synergize with CaO or LiOH to promote CO_2 fixation as $NaHCO_3$ and reduce total energy consumption (Dindi et al., 2015; Lee et al., 2021). In chemical precipitation method, MEA has been used alongside Na_2SO_4 to selectively remove Ca^{2+} as $CaSO_4$, followed by the formation of high-purity carbonates through controlled reactions with absorbed CO_2 (Kang et al., 2018; Kim et al., 2020).

Experimental evidence supports the effectiveness of these hybrid approaches. Liu et al. (2014) demonstrated that MEA, when paired with seawater effluent, could produce high-purity $CaCO_3$. Wang et al. (2015) further showed that the pH buffering capacity of desalination brine enhances the rate of CO_2 absorption. Ho and Iizuka (2023) proposed using carbide slag as a regenerant for spent or acidified amines, enabling multiple carbonation cycles without significant performance loss. Across studies, typical CO_2 absorption capacities exceed 0.9 mol CO_2/mol amine, with carbonate products achieving purities above 90% (Yoo et al., 2022).

Despite its advantages, widespread adoption of chemical absorption methods remains limited by the high energy requirements for solvent regeneration, potential amine degradation, and the need for corrosion-resistant equipment. To mitigate these drawbacks, ongoing research focuses on developing more stable and selective absorbents, incorporating solid-supported hybrid method, and exploring low-energy or renewable-powered regeneration schemes (Ho and Iizuka, 2023).

3 Method comparison and technical evaluation

The four primary pathways of seawater-based mineral carbonation—namely the pH swing method, the modified Solvay process, the chemical precipitation method, and the chemical absorption method—present distinct mechanisms, operational profiles, and technological readiness levels. This section offers a systematic comparison based on technical principles, process complexity, CO_2 capture

Table 2. Comparison of four methods for seawater-based carbon mineralization: process conditions, economic costs, industrial demonstration, challenges, and safety considerations

Method	Mechanistic complexity	CO_2 capture efficiency/%	Economic cost	Industrial demonstration	Problems and solutions encountered	Integration/application scenario
pH swing method	CO_2 is dissolved and mineralized into calcium/magnesium carbonate by adjusting pH; alkaline substances like NaOH and NH_4Cl are used; pH and temperature are optimized	44.3–91.3	the primary cost is the consumption of chemicals, such as NaOH; this can be reduced through electrolysis	the industrial demonstration is in its early stage, with promising results in laboratory conditions	(1) high chemical consumption: reduce consumption by producing NaOH *in situ* via electrolysis; (2) ion competition: Optimize pH control to mitigate competition	modular systems: aquaculture, offshore units, and cooling water outlets
Modified Solvay process	use of CaO or $Ca(OH)_2$ instead of NH_3; high-salinity seawater or wastewater as a raw material; optimization of temperature and pressure conditions	>90	lower costs by replacing the alkali source and improving KOH concentration and temperature to optimize energy efficiency	good performance in pilot-scale trials, with certain commercialization potential	(1) chloride ion interference: develop new adsorbents to remove chloride ions and improve absorption efficiency; (2) high equipment requirements: efficient equipment is needed to support large-scale applications	centralized: desalination plants, industrial brine systems
Chemical precipitation method	use reagents like Na_2SO_4 to separate Ca^{2+} and Mg^{2+}, and enhance CO_2 capture efficiency through pH adjustment or the addition of specific absorbents	33.2–92	reagent costs (Na_2SO_4, NaOH) offset by high-purity $CaCO_3$/$MgCO_3$ products; cost-effective if NaOH is sourced via seawater electrolysis	already applied on a certain scale, particularly suitable for large-scale industrial use	(1) precipitation separation issues: use Na_2SO_4 to separate Ca^{2+} and Mg^{2+} to improve product purity; (2) raw material selection: control ion concentration in seawater and wastewater to avoid side reactions	well-suited for large-scale coastal industries, offering low-cost, high-purity carbonate recovery with simple operation
Chemical absorption method	use alkanolamine absorbents (e.g., MEA, AMP, MDEA, PZ) to capture CO_2, followed by reaction with Ca^{2+} and Mg^{2+} in seawater or wastewater; influenced by absorbent structure and pH conditions	Up to 95	higher chemical costs are due to expensive absorbents; operational costs depend on absorbent regeneration efficiency and energy input	mainly at laboratory and small pilot scales; further development is needed for the engineering scale-up	(1) absorbent regeneration: improve regeneration efficiency to reduce operational costs; (2) absorption performance: Optimize amine structure and system design for better CO_2 uptake and selectivity; (3) compatibility with seawater ions and waste brine must be ensured to avoid inhibition or undesired reactions	point sources (e.g., flue gas), paired with seawater treatment and pH control

efficiency, material requirements, and scalability, providing insights into potential application scenarios and optimization strategies (Table 2).

3.1 *Process characteristics and mechanistic complexity*

The pH swing method is characterized by its relatively simple conceptual design and operational flexibility. The method enables selective precipitation of Ca^{2+} and Mg^{2+} by pH modulation, often through NaOH addition or *in situ* electrochemical generation. However, the method requires precise pH control to avoid unintended co-precipitation and loss of efficiency, especially in Mg^{2+}-rich systems. Integration with electrolysis and waste-derived alkalis has reduced chemical costs and broadened application potential (Wang et al., 2011; Choi et al., 2021). The modified Solvay process involves multi-step operations including alkali treatment, CO_2 absorption, precipitation, and product recovery. It benefits from compatibility with brine-rich environments, offering simultaneous CO_2 fixation and ion removal. However, its reliance on temperature control, pressure maintenance, and specialized reactors increases technical complexity and infrastructure demand (El-Naas et al., 2017; Lee et al., 2021). The chemical precipitation method employs a simple and controllable two-step process, consisting of ion separation followed by carbonate precipitation. This approach is operationally straightforward. However, its CO_2 capture efficiency is often limited, unless supported by pH adjustment or the addition of specific absorbents (Kim et al., 2020; Ho and Iizuka, 2023). Moreover, improving the selectivity of ion precipitation and enhancing the purity of the resulting carbonate products remain key challenges. The chemical absorption method utilizes amine-based solvents to capture CO_2, followed by mineralization through reaction with Ca^{2+}/Mg^{2+} in seawater. While this route allows for efficient gas-phase CO_2 removal and high carbonate yields, it also requires solvent regeneration, pH buffering, and careful handling due to amine volatility and degradation risk. The chemical absorption method is most effective when integrated into other systems, such as pH swing or precipitation-based processes. It enhances gas-phase CO_2 capture, buffers pH, and supports downstream carbonate precipitation, complementing and improving the overall system performance (Kang et al., 2018; Yoo et al., 2022).

3.2 *Technical maturity and CO_2 capture performance*

The pH swing method has reached pilot-scale demonstration, with documented CO_2 capture efficiency up to 44.3%–91.3% using desalination brine and seawater-basedwastewater(Kangetal.,2020;Choietal.,2021;Choetal., 2022). Its relatively low barrier to implementation makes it suitable for decentralized applications such as aquaculture, offshore platforms, and coastal treatment units. The modified Solvay process has been validated in bench-scale systems. It is particularly promising in desalination plant integration, where high-salinity brine serves as both reactant and solvent. CO_2 capture efficiency over 90% and Na^+ removal rates of 35%–45% have been reported, along with multiple product streams (e.g., $NaHCO_3$, $MgCO_3 \cdot 3H_2O$) (El-Naas et al., 2017; Lee et al., 2021). The chemical precipitation method achieves high CO_2 capture efficiency (up to 92%) and product purity (>90%) in seawater-based wastewater. Using NaOH and Na_2SO_4 enables selective Ca^{2+} removal and promotes co-precipitation of $MgCO_3$ and $NaHCO_3$ (Kang et al., 2018; Ho and Iizuka, 2023). The chemical absorption method achieves high CO_2 capture efficiency (up to 95%) and produces high-purity carbonates using regenerable amines and seawater-based solutions (Liu et al., 2014; Kang et al., 2017). Long-term viability depends on improving amine regeneration techniques and solvent durability.

3.3 *Economic and operational considerations*

Beyond technical feasibility, the economic and operational aspects of each pathway critically influence their large-scale deployment potential. Key factors include chemical input costs, process complexity, energy consumption, byproduct value, environmental safety, and scalability. The pH swing method generally requires significant alkali input (e.g., NaOH). However, this can be mitigated through in situ electrochemical generation or the reuse of industrial alkaline waste such as steel slag or cement kiln dust (Wang et al., 2011; Zhang et al., 2019). The modified Solvay process, while offering valuable byproducts and integration potential with brine-rich systems, involves multistage operations with significant energy and capital demands (El-Naas et al., 2017; Lee et al., 2021). The chemical precipitation method primarily incurs costs from the use of reagents such as Na_2SO_4 and NaOH. Nevertheless, it generates high-purity byproducts like $CaCO_3$ and $MgCO_3$, which offer substantial economic value due to their enhanced purity (Ho and Iizuka, 2023). Furthermore, as NaOH can be produced via seawater electrolysis, this approach remains both cost-effective and practical, enabling efficient conversion of CO_2 into valuable metal carbonates. In contrast, the chemical absorption method is highly efficient but energy- and solvent-intensive, with environmental concerns related to amine degradation and toxicity (Kang et al., 2017). As such, its role as an auxiliary pathway in hybrid systems can offset operational burdens while maximizing CO_2 capture and product selectivity.

3.4 *Integration potential and application scenarios*

The pH swing method is well-suited for modular, distributed applications with access to renewable electricity and waste alkalis. They can be deployed at aquaculture farms, offshore carbon farms, or industrial cooling water

outlets. Modified Solvay process are optimal for centralized industrial systems, especially where desalination or brine discharge occurs. The ability to co-recover multiple solid products adds economic incentive and supports closed-loop water-carbon cycles. The chemical precipitation method is ideal for large-scale coastal or industrial applications with access to seawater or brine. Its simple operation and ability to produce high-purity carbonate products make it suitable for rapid deployment and resource recovery. Integration with electrochemical pH control can further enhance CO_2 capture efficiency and sustainability. Amine-based absorption method is promising for high-concentration CO_2 point sources (e.g., flue gas), particularly when combined with seawater pretreatment units and pH regulation. Further development of low-cost, low-toxicity absorbents is essential to improve sustainability and reduce lifecycle emissions.

4 Potential for large-scale application of seawater-based solution mineral carbonation technology

Given the abundance of seawater resources and the large volumes of seawater-based solutions from desalination brine and seawater-based wastewater, using seawater-based solutions for mineral carbonation holds significant development potential. Compared to traditional carbonation technologies, seawater-based solution carbonation offers low cost, abundant resources, and minimal environmental impact (Huijgen et al., 2007; Ukwattage and Ranjith, 2018). Unlike traditional chemical solvents, seawater-based solutions reduce costs and offer an environmentally friendly solution as a widely available resource (Ho et al., 2021). Furthermore, this technology not only captures atmospheric CO_2 but also generates additional benefits, such as waste management and the utilization of carbonation products, thereby enhancing the economic feasibility of the technology (Ho et al., 2023; Ho and Iizuka, 2023). For example, carbonation products are considered for use as cementing materials (Pan et al., 2016; Ho et al., 2021), soil stabilizers (Ukwattage and Ranjith, 2018), and plant growth additives (He et al., 2017). Industrially, they can serve as cementing materials, paper fillers, acid reducers, fertilizers, paint fillers, and chemical raw materials (Woodall et al., 2019; Lee and Lee, 2021). Moreover, compared to other CO_2 sequestration and utilization technologies, even at low CO_2 concentrations and atmospheric pressure, high-salinity desalination water and seawater-based wastewater can react directly with various flue gases to fix CO_2. This can avoid costly, energy-intensive CO_2 purification and pressurization processes (Ho and Iizuka, 2023). Additionally, compared to using natural seawater for mineral carbonation, the higher concentrations of calcium and magnesium ions in seawater-based solutions lead to more efficient carbonation and higher purity of carbonation products (Ho and Iizuka, 2023).

However, the implementation of this technology also faces several challenges. Firstly, although the pH swing method can effectively control the conditions of the CO_2 carbonation process, it also faces issues related to chemical consumption and waste production. A deeper exploration of its large-scale operation and commercial potential is needed to optimize the entire process, including techno-economic assessments and material flow analyses. The modified Solvay process, which integrates alternative alkaline substances, optimized adsorbents, and multistage indirect carbonation, shows promise for enhancing CO_2 capture efficiency and converting CO_2 into value-added compounds. However, this process requires safe, efficient equipment and precise experimental conditions. Chemical precipitation and chemical absorption methods both enhance CO_2 fixation efficiency and promote the production of high-purity carbonate minerals; however, the costs of chemical consumption, as well as the scalability and engineering implementation of the systems, must be considered. Finally, economic analysis is also a critical issue, and a more comprehensive techno-economic assessment is needed to demonstrate its commercial viability (Ho and Iizuka, 2023). Technology integration is also a significant challenge, as integrating seawater carbonation technology with existing industrial processes is necessary to optimize and efficiently utilize resources (Ho and Iizuka, 2023). Moreover, market demand and application potential for carbonation products require further research (Ho and Iizuka, 2023). Finally, byproducts from carbonation, such as acidic wastewater and chlorides, need appropriate treatment to ensure environmental safety (Liu et al., 2021; Duan et al., 2023).

5 Conclusions and outlook

This article has explored the concept, mechanisms, and current research progress of capturing CO_2 through mineral carbonation using seawater-based solutions. As an effective CO_2 capture strategy, seawater mineral carbonation technology offers the advantage of reducing atmospheric CO_2 concentrations at low cost and generates additional economic benefits, demonstrating good technical feasibility. Compared to traditional carbonation techniques, this technology leverages the alkalinity of seawater and its abundant Ca^{2+} and Mg^{2+} to achieve stable transformation of CO_2, possessing significant market potential and lower energy input (Table 3). However, several challenges arise during the implementation process, including chemical consumption and waste management issues associated with the pH swing method, as well as the need for precise conditions and efficient, safe equipment in methods such as the modified Solvay process. Additionally, techno-economic assessments, material flow analysis, and the scalability and engineering implementation of the system need to be thoroughly considered. Economic analysis is crucial for verifying its commercial viability, while market demand for carbonation products, application po-

Table 3. Comparison of seawater-based mineral carbonation and traditional CO_2 capture methods

Aspect	Seawater-based mineral carbonation	Traditional mineral carbonation technology
Resource and economic feasibility	utilizes abundant seawater with low resource cost and initial investment; further techno-economic assessment is needed	relies on specialized chemicals or minerals; higher initial cost, but stable due to mature technology
Environmental and byproduct impact	environmentally friendly with reduced harmful emissions; requires treatment of acidic wastewater and chlorides to ensure safety	it may generate more waste and pollutants, and byproduct management is more challenging.
Technical maturity and integration	relatively simple, especially with the ph swing method; requires integration with existing industrial systems for optimization	technologically mature but complex; may require system upgrades for improved efficiency
CO_2 capture and conversion efficiency	capable of reacting directly with low-concentration CO_2 (e.g., flue gas); high efficiency and product purity ($CaCO_3$, $MgCO_3$)	requires high CO_2 concentration and purification; generally lower efficiency and product purity
Application potential and scalability	carbonates can be used in cement, soil stabilizers, fertilizers, etc.; large-scale deployment potential, but commercialization challenges remain	fewer industrial applications; scalable due to maturity, but constrained by high energy demand and cost

tential, and environmental safety management of byproducts are key factors that cannot be overlooked in implementation. Moreover, further research and analysis are needed to apply carbonation products, particularly as life cycle assessments and techno-economic analyses are currently lacking. Although seawater-based solutions are abundant, the application potential, value, and demand for carbonation products will be limiting factors in the technology's development. These assessments will help to formulate development strategies for seawater-based mineral carbonation and establish methods for calculating CO_2 reduction. Different boundary conditions, such as various types of seawater wastewater, carbonation products and byproducts, and energy utilization, are crucial for assessing the actual CO_2 reduction. Additionally, product development is essential to enhancing the feasibility of the technology and its integration with other systems. Given the high content of Ca^{2+} and Mg^{2+} in seawater and its diversity, a variety of products can be developed through different carbonation methods. Exploring applications for these products can enhance economic benefits and increase the feasibility of the technology. At the same time, addressing residual acidic wastewater and byproducts, such as chlorides and sulfates, is necessary to ensure environmental safety, enhance the feasibility of the technology, and achieve sustainable development goals.

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